

# HAT-P-32b

R-0161

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-32\_231012 · created 2026-07-16T15:49:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460229.7678 +/- 0.0017 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1409 +/- 0.0053               |
| Transit depth (Rp/Rs)^2                           | 1.98 +/- 0.15 [%]               |
| Orbital Inclination (inc)                         | 89.53 +/- 1.07                  |
| Airmass coefficient 1 (a1)                        | 1.2656 +/- 0.0088               |
| Airmass coefficient 2 (a2)                        | -0.1773 +/- 0.005               |
| Scatter in the residuals of the lightcurve fit is | 0.69 %                          |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 4.65                            |
| Optimal Annulus                                   | 8.92                            |
| Transit Duration (day)                            | 0.12932 +/- 0.00096             |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                          |
|--------------------------------------|----------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.1409 $\pm$ 0.0053 vs 0.1489 (deviation 1.51 $\sigma$ ) |
| Duration fit vs published            | 0.12932 d vs 0.1304 d (deviation 1.14 $\sigma$ )         |
| Mid-time O-C                         | -1.8 min                                                 |
| Depth SNR                            | 26.6                                                     |
| Residual scatter                     | 0.69 %                                                   |

## Light curve

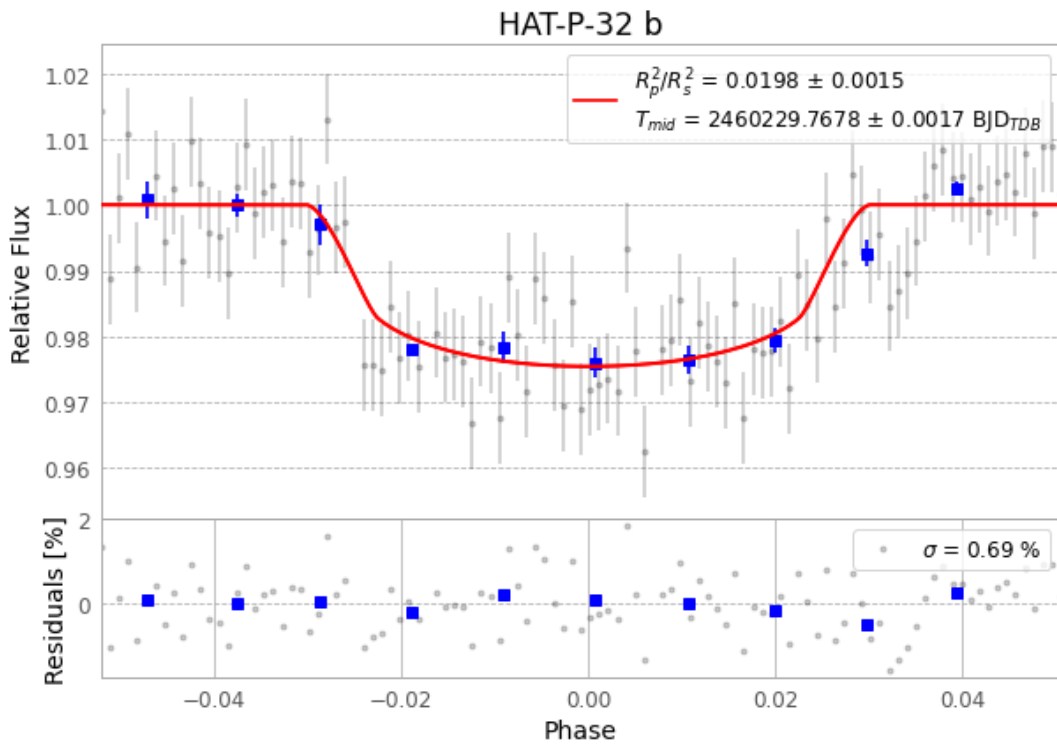


Figure 1 — FinalLightCurve\_HAT-P-32 b\_2023-10-11.png (SHA-256 77f00843caf7a6c0...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 235], 2 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 bb03b0eb59488a75...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1634555

## Data provenance

107 FITS frames (70 MB), HATP-32231012033615.FITS → HATP-32231012085415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-231012.log (e1493231c023...), FinalLightCurve\_HAT-P-32 b\_2023-10-11.png (77f00843caf7...), FinalLightCurve\_HAT-P-32 b\_2023-10-11.csv (0210c9d34d60...)

## Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 15:49Z.